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ADVERSE CHILDHOOD EXPERIENCES & POST-TRAUMATIC GROWTH PREDICT DESTRUCTIVE SEXUAL BEHAVIOR IN COLLEGE STUDENTS

NASTAJIA K. HAMILTON & DONNA W. NELSON

WINTHROP UNIVERSITY

Abstract – Our research examined whether suffering from sexual or domestic violence and/or other emotional and physical trauma during childhood (known as adverse childhood experiences) leads to unhealthy or destructive sexual behavior in adulthood. The goal was to replicate findings showing that adverse childhood experiences (ACEs) increase risky sexual behavior, and to extend this work to examine whether ACEs place a person at greater risk of hypersexuality. We also examined the extent to which having strong social support and experiencing post-traumatic growth (PTG) mitigated some of the negative effects of ACEs. Results indicated that experiencing ACEs predicted more hypersexuality and risky sexual behavior, while PTG predicted less hypersexuality. Social support did not predict hypersexuality or risky sexual behavior, although it was associated with greater PTG. These findings increase our understanding of the impact of ACEs and highlight protective factors, suggesting potential avenues for mitigating some of the devastating effects of adverse childhood experiences.

Although adverse childhood experiences (ACEs) are no new concept, in recent years, their prevalence and wide impact has been noted. In fact, according to the Centers for Disease Control, around 61% of adults surveyed across the U.S. reported they had experienced at least one type of ACE, and about one in six reported they had experienced four or more types of ACEs (CDC, 2022). By definition, ACEs are potentially traumatic events that occur prior to the age of 18 that can have long-term, detrimental consequences for one's health and well-being in adulthood (Chang et al., 2019). Examples include violence, abuse, neglect, witness to violence in the home or community, substance abuse issues, mental health issues, and instability due to parental separation (Baglivio et al., 2017).

Recent research conducted on the immediate or short-term effects of ACEs has shown that the stress and trauma associated with ACEs can cause children to experience anxiety, depression, aggression, learning impairments, and decreased immune function (Maroney, 2020). Other research has shown that ACEs can have a variety of long-term effects throughout adulthood including premature death, increased comorbid conditions, chronic diseases, such as cancer, and risk behaviors, such as smoking and substance abuse (Chang et al., 2019; Felitti et al., 1998). Furthermore, ACEs have been linked to risky sexual behaviors, such as having multiple sexual partners, participating in unprotected sex, anal intercourse, first engaging in sexual intercourse at an early age, having sex under the influence of drugs and/or alcohol, and commercial sex (Akumiah et al., 2020).

The aforementioned research confirms that ACEs have a variety of significant, detrimental consequences that may persist throughout the lifespan. In particular, early trauma has been linked with risks to emotional and physical well-being as well as high rates of dysfunctional behavior. Given these findings, we wondered whether ACEs would also be associated with greater risk of hypersexuality (i.e., compulsive sexual thoughts and behavior). Hypersexual individuals experience preoccupation with intense sexual fantasies, desires, or urges and compulsive sexual actions that are difficult to control, cause discomfort, or have a detrimental impact (Kellett et al., 2017). Hypersexual behaviors tend to persist and solidify into patterns due to their propensity to regulate internal mental conditions, such as reducing anxiety or enhancing excitement (Montaldi, 2002). It is possible that hypersexuality functions as a method of coping with adversity and therefore, we expected to observe higher rates of hypersexuality in adults who experienced more early traumatic events.

A second research question we explored concerns potential protective variables that may enhance coping with ACEs. A robust literature indicates that more successful coping with early trauma is associated with strong support from one's social network (Tucker et al., 2020). For example, perceived social support from family and friends has been shown to minimize the damaging effects of adverse events and promote healthy functioning (Cohen & Willis, 1985; Taylor, 2011). In contrast, low levels of social support from family and friends has been linked to isolation, anxiety, misconduct, and low self-esteem (Gariépy et al., 2016; Hartup & Stevens, 1997). Furthermore, according to Lester et al. (2020), social support from peers enhances coping with ACEs by creating a beneficial sense of relatedness and solidarity. In regard to support from community members, nonparental adults serving as mentors promote coping by offering tangible aid, imparting knowledge and skills, and increasing self-esteem of children who experienced ACEs (Southwick et al., 2006).

Given the potential beneficial effects of coping resources for those who experience early trauma, investigation of other variables that may serve a protective role is warranted. One promising line of inquiry is examining the link between post-traumatic growth (PTG) and coping with ACEs. PTG is conceptualized as the propensity of individuals to function at a healthier level in the aftermath of traumatic events, as a result of discovering meaning in their traumatic experiences (Tedeschi & Calhoun, 1996). PTG is demonstrated by a greater appreciation for life, better relationships, improved sense of personal strength, and even spiritual growth (Tedeschi & Calhoun, 1996). It is possible that PTG may mitigate some of the harmful long-term consequences of ACEs. Specifically, if individuals are able to process their trauma in an adaptive way and find meaning in their suffering, they may eventually be equipped to move beyond prior adversities by leveraging their experiences in a beneficial way. This notion poses possible solutions for the mitigation of symptoms stemming from childhood trauma and other traumatic events throughout life and will be further explored in our study.

The aim of our research was to examine links between ACEs, destructive sexual behavior, social support and PTG. We hypothesized that experiencing ACEs would be associated with more risky sexual behavior and higher rates of hypersexuality. We also hypothesized that having strong social support and experiencing PTG would be associated with lower rates of risky sexual behavior and less hypersexuality.

Method

Participants

An anonymous online survey was completed by 102 participants. Participants were recruited from college classrooms, social media, and personal contact. The gender distribution was as follows: 14% were men, 78% were women, one percent were transgender women, six percent were non-binary, and two percent of individuals chose not to identify their gender. The race/ethnicity distribution was as follows: 60% of participants were Caucasian, 33% were African-American, and three percent were Asian or Asian Indian, one percent were American Indian or Alaska Native, two percent reported other, and one percent chose not to identify their race/ethnicity. The mean age was 20 (*SD* = 2.90). In addition, 62% of participants identified as heterosexual, six percent were homosexual, 22% were bisexual, nice percent identified as other, and two percent of individuals chose not to identify their sexuality. Some participants were given extra course credit for participation. All participation was voluntary, and the project was deemed as exempt from oversight by the University Institutional Review Board.

Procedure

Interested participants were directed to an online system where they first encountered the informed consent form. Continuing into the survey was taken as an indication of consent. Participants had the option to exit the survey at any time, at which point they would see a standardized page thanking them for their participation.

Materials

Adverse Childhood Experiences

To begin, participants were asked to complete the BRFSS Adverse Childhood Experience (ACE) Module (US Centers for Disease Control and Prevention (CDC), 2019). This scale is used to determine the extent to which individuals experienced different adverse childhood experiences during the time period prior to 18 years of age. A total of 11 possible adverse events were assessed. Examples included living with a mentally ill family member, experiencing parental separation, experiencing emotional or physical abuse and living with a family member who abused substances. Respondents indicated either "yes" or "no" to indicate whether or not they experienced the event listed for each item. The published Cronbach's Alpha for this measure is .78 and we obtained a Cronbach's Alpha of .78.

Post-Traumatic Growth

Next, participants were asked to complete the Post-Traumatic Growth Inventory (Tedeschi & Calhoun,

1996). This scale is used to determine the extent to which individuals who experienced adverse events report beneficial changes that occurred in their life as a result of those experiences. They indicated the extent to which they experienced 21 possible areas of growth and change such as "I have a greater appreciation for the value of my own life", "I discovered that I'm stronger than I thought I was", and "I have a greater feeling of self-reliance". Responses were made on a Likert scale ranging from zero (not at all) to five (to a very great degree). The published Cronbach's Alpha for this measure is .90 and we obtained a Cronbach's Alpha of .92.

Social Support

Then, participants responded to a measure designed to determine perceptions of current social support through the Multidimensional Scale of Perceived Social Support (Zimet et al., 1988). This scale consists of three subscales, each with four items, that measure three facets of social support. The first subscale asked participants about their perceived support from family, the second asked participants about their perceived support from friends, and the third asked participants about their perceived support from a significant other. Participants indicated their agreement with statements such as "I get the emotional support I need from my family", "I have friends with whom I can share my joys and sorrows", and "I have a special person who is a real source of comfort to me". Responses were made on a Likert scale ranging from one (very strongly disagree) to seven (very strongly agree). The published Cronbach's Alpha for this measure is .88 and we obtained a Cronbach's Alpha of .89.

Hypersexuality

Next, participants responded to a scale measuring engagement in hypersexual behaviors known as the Hypersexual Behavior Inventory (Reid et al., 2011). They indicated the extent to which they have displayed 19 possible patterns of hypersexual behavior such as "I engage in sexual activities that I know I will later regret", "When I feel restless, I turn to sex in order to soothe myself", and "Sex provides a way for me to deal with the emotional pain I feel". Responses were made on a Likert scale ranging from one (never) to five (very often). The published Cronbach's Alpha for this measure is .95 and we obtained a Cronbach's Alpha of .93.

Risky Sexual Behavior

Then, participants were asked to respond to the Sexual Risk Behavior Scale (Fino et al., 2021). This six-item scale is designed to measure the extent to which individuals participate in sexually risky behaviors or have poor sexual health practices. Respondents indicated the extent to which they engaged in risky sexual behaviors including vaginal, anal and/or oral sex without a condom, sex while under the influence of alcohol or drugs, and unprotected sex with strangers. Responses were made on a Likert scale ranging from one (never) to five (very often). The published Cronbach's Alpha for this measure ranges from .76 to .84 and we obtained a Cronbach's Alpha of .77.

Finally, participants responded to commonly used demographic items including age, race, gender, socioeconomic status, and parental education level.

Results

Means were calculated for each measure for all participants. The average ACE score in our sample was 3.05 (SD = 2.54). The overall mean PTG score was 3.38 (SD = .74). The overall mean Social Support score was 5.33 (SD = .97). The overall mean Hypersexuality score was 1.57 (SD = .64) and the overall Risky Sexual Behavior score was 1.78 (SD = .81). No differences in Hypersexuality or Risky Sexual behavior were found across any of the demographic variables and thus, these variables were not included in subsequent analyses.

We conducted a multiple regression to examine the combined effects of ACEs, PTG and social support in predicting risky sexual behavior. The overall model was a good fit [F (3, 95) = 3.30, p = .024]. Furthermore, ACEs contributed significantly to the model and accounted for a unique portion of the variance in risky sexual behavior scores (β = .31, p < .01), while both PTG (β = -.04, p = .695) and social support (β = .13, p = .16) did not. Next, we conducted a multiple regression to examine the combined effects of ACEs, PTG and social support in predicting hypersexuality scores. The overall model was a good fit [F (3, 92) = 8.52, p < .001]. Furthermore, ACEs contributed significantly to the model and accounted for a unique portion of the variance in hypersexuality (β = .377, p < .01). PTG also contributed significantly to the model and explained a unique portion of variance in hypersexuality (β = -.28, p < .01), while social support did not (β = .112, p = .27).

We conducted additional analyses to test the relationship between social support and PTG. Pearson's correlations yielded a positive association between these variables [r (101) = .39, p < .001]. Those with strong social support also reported more beneficial psychological changes as a result of experiencing trauma.

Discussion

As has been documented in prior research (e.g., Green et al., 2005; Hillis et al., 2001), we hypothesized that experiencing more ACEs would predict more risky sexual behavior. We also expected ACEs to predict

greater hypersexuality. Both of these hypotheses were supported. Replicating prior findings, participants in our sample who reported more adverse events in childhood were also more likely to report engaging in risky sexual behaviors ranging from having sex while under the influence of drugs and/or alcohol to participating in unprotected sex with strangers. Extending this work, we also found that those who experienced more ACEs were more likely to experience hypersexuality, involving excessive or uncontrollable sexual thoughts, urges, or behaviors. One possible explanation for the link between ACEs and hypersexuality is that childhood trauma can disrupt the development of healthy attachment and bonding between caregivers and children (Shonkoff & Garner, 2012). This can have a variety of detrimental effects, such as difficulty regulating or controlling one's emotions and coping with stress (Anda et al., 2006). Therefore, hypersexuality can be a way of seeking intimacy, pleasure, or a sense of control, and may serve as a temporary escape from the negative emotions associated with trauma.

We also hypothesized that PTG would emerge as a protective variable, reducing the risk of risky sexual behavior and hypersexuality. Our findings provided partial support for this hypothesis. As expected, PTG emerged as a protective variable in relation to hypersexuality. Essentially, those who reported undergoing significant and beneficial psychological changes in the aftermath of their trauma were less likely to report engaging in excessive or compulsive sexual thoughts, feelings, and behaviors. However, PTG did not emerge as a protective variable for risky sexual behavior. One possible reason for these differential findings could be that hypersexuality and risky sexual behavior represent different patterns of sexual behavior. Hypersexuality is characterized by overwhelming sexual fantasies, desires, and conduct, while risky sexual behavior involves engaging in sexual activities that have the potential to result in negative consequences such as sexually transmitted diseases or unwanted pregnancy. It is possible that PTG may be more protective against hypersexuality than it is of risky sexual behavior because hypersexuality may be a coping mechanism for individuals who have experienced trauma, whereas risky sexual behavior may reflect other underlying factors such as impulsivity or substance use. In other words, those who have experienced trauma may become hypersexual as a way of coping with feelings of anxiety or depression while risky sexual behavior may be a sign of more impulsive or sensation-seeking inclinations.

Furthermore, we hypothesized that social support would operate as a protective variable with respect to risky sexual behavior and hypersexuality. Our results did not confirm this hypothesis, as social support did not emerge as a significant predictor of either risky sexual behavior, or hypersexuality, after controlling for ACEs. This suggests that destructive sexual patterns of behavior are more strongly influenced by early childhood trauma and less so by having strong social support in adulthood. These findings are consistent with research and theory that highlights the power of early childhood events to shape the future lives of individuals, including a variety of emotional, physical, and social impacts, such as distress, illness, and anxiety (Davis et al., 2002; Chang et al., 2019; Maroney, 2020). It is possible that had we measured social support experienced in childhood rather than currently in adulthood, this variable may have been more influential.

Finally, we found evidence that social support and PTG were positively correlated. This suggests that social support may play an indirect role in preventing negative long-term effects of ACEs. Perhaps receiving social support from home, school, teachers, or mentors can help facilitate post-traumatic growth, thereby teaching people how to process their trauma in more productive ways. This could also help reduce the risk of negative outcomes such as destructive sexual behavior later in life. Future research should explore these possibilities.

Our study had limitations. To begin, our sample lacked diversity on a variety of levels, differing from the broader population of college students in some ways. As the demographics of the sample show, more than three-fourths of the sample consisted of women (78%), while in the broader population of U.S. college students, approximately 60% are women (West, 2021). Similarly, while over half of our participants identified themselves as White (60%), about 52% of the broader college population in the U.S. identify as White (Barshay, 2023). However, the demographic numbers in our sample are in line with the proportions of demographic categories found at the university from which many of the participants were recruited. Another limitation concerns the measure of social support utilized in our study. Specifically, the scale only measured current, perceived social support, excluding received social support or social support in childhood as potential predictive variables. Our findings would be more nuanced if we had measured both components of social support and tested their unique significance at different points in the lifespan.

This study also gave rise to ideas for future research. Future researchers may want to measure and compare the impacts of social support received during childhood in the aftermath of ACEs versus social support

received in adulthood to see whether it operates as a more influential protective variable when experienced during childhood. Similarly, future researchers may want to measure and compare the significance of PTG achieved during childhood in the aftermath of ACEs versus PTG achieved in adulthood to see whether its protective effects are more prominent during childhood. With a larger, more diverse sample, researchers could also examine whether race, ethnic background, culture, or religion predict the level and impact of perceived social support for survivors of childhood trauma. For example, do customs and beliefs influence individuals' sense of community and support, especially in the context of such a sensitive topic? In addition, future research could specifically compare the influence of received social support and perceived social support on patterns of hypersexuality in the wake of ACEs. Lastly, researchers could examine whether religiosity predicts the degree to which individuals with ACEs experience hypersexuality and achieve PTG. For example, does religious conviction impact susceptibility to hypersexual behavior, and does faith in a higher power give individuals a greater sense of purpose regardless of the traumas they faced in adolescence?

The findings of our study increase our understanding of the impact of ACEs and potential protective factors. Our findings also suggest potential avenues for mitigating some of the devastating effects of adverse childhood experiences, such as developing PTG interventions to promote beneficial psychological changes following trauma. Discovering ways to reduce victims' risk of developing hypersexuality can have significant mental health implications because those who experience hypersexuality are often further traumatized in adulthood due to social rejection, disapproval, stigmatization, degradation, and negative evaluation (Reid et al., 2012). On the bright side, by addressing and mitigating the negative impacts of ACEs, these unnecessary, diminishing and destructive effects surrounding hypersexuality can be avoided and survivors of childhood trauma can heal and find peace in looking towards a better, more promising future.

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Author Note
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Correspondence concerning this article should be addressed to Nastajia Hamilton at hamiltonn2@winthropalumni.edu